



Bolla Lokesh Reddy

B.Tech Computer Science and Engineering

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Degree	University/Institute	Year	CGPA/Marks(%)
B.Tech Computer Science and Engineering	IIT Hyderabad	2027	8.3
XII (Board of Intermediate Education, Andhra Pradesh)	Sri Chaitanya Junior College, Guntur	2023	9.57
X (Board of Secondary Education, Andhra Pradesh)	Bhashyam Educational Institution, Guntur	2021	10.00

SCHOLASTIC ACHIEVEMENTS

- Secured AIR **953** in **JEE Advanced 2023 (Top 0.5%)**.
- Secured AIR **1541** in **JEE MAINS 2023 (Top 0.15 %)**.
- Secured State Rank **118** in **TS EAMCET 2023 (Top 0.05%)**.
- Secured State Rank **257** in **AP EAPCET 2023 (Top 0.1%)**.

PROJECTS

Infinite-Precision Arithmetic | C++ (Coursework)

- Implemented arithmetic operations (add, sub, mult, div) for arbitrarily large integers and floating-point numbers.
- Used string manipulation and dynamic memory handling to overcome language limitations.

Multithreaded Sudoku Validator | C++ (Coursework)

- Used TAS, CAS, and Bounded CAS for concurrency control and validated Sudoku grids in parallel.
- Measured execution performance over varying threads and grid sizes.

Semaphore Synchronization — OS (Coursework)

- Implemented C++ solutions for Fair Readers-Writers and Writers Preference problems using semaphores.
- Conducted worst-case and average access time analysis.

RISC-V Simulator | C++ (Coursework)

- Built a custom simulator to decode and execute RV32I instruction set with memory and register modules.
- Enabled debugging via CLI interface and memory inspection features

Cache Simulator (Coursework)

- Designed and evaluated a CPU cache simulator in C++ with support for direct-mapped, set-associative, and fully-associative configurations.
- Benchmarked hit/miss ratios under multiple policies

Food Image Classifier | Python, TensorFlow, Keras

- Developed a CNN model to classify food images into 10 categories.
- Applied data augmentation and explored pretrained models like ResNet for accuracy improvement

Helmet Detection using YOLOv8 | Python, PyTorch

- Trained a YOLOv8 model to detect helmet violations in traffic scenes using bounding box annotations.
- Fine-tuned with transfer learning and visualized results with real-time detection support

GraphX — Graph Domain-Specific Language

Team of 6

C++, Compiler Design

- Designed a **domain-specific language (DSL)** for **graph programming** with first-class graph, vertex-set, and edge-set abstractions, enabling high-level construction and manipulation of dynamic graphs.
- Defined graph-specific operators and built-in algorithms including **BFS/DFS, Dijkstra, Bellman-Ford, MST, topological sorting, connected components, bridges, and articulation points**.
- Developed **density-aware graph optimizations** that select adjacency-list, adjacency-matrix, or hybrid representations based on graph density to improve storage and algorithmic efficiency.

Document Management Repository (DMR)

Team of 6

React, Node.js, Express.js, MongoDB, Redis, AWS S3, Groq LLaMA | Software Engineering

- Developed a full-stack document management platform supporting **AI-powered document tagging and intelligent vault routing** using Groq LLaMA, alongside manual organization and document lifecycle management.
- Built a REST-based architecture using **React, Node.js, Express.js, and MongoDB**, with **AWS S3** for streaming file storage and Redis caching for frequently accessed resources.
- Implemented secure document and workspace management with **JWT authentication, role-based access control, developer API keys, granular permissions, storage quotas, audit functionality, and soft-delete/recovery workflows**.

Distributed Bitcoin Mining Framework

Team of 2

C++, MPI, Pthreads, OpenSSL | Distributed Computing

- Designed a Bitcoin mining framework implementing **static multithreaded partitioning, dynamic work stealing, and MPI-based distributed mining** for parallel nonce search using SHA-256 hashing.
- Implemented dynamic work distribution and parallel execution across threads and processes, addressing load imbalance in fixed-partition mining through fine-grained work allocation.
- Developed a distributed mining architecture with **Raft-based leader election, checkpointing, and fault-recovery mechanisms**, enabling resilient multi-node execution.
- Benchmarked the three execution models across varying **thread/process counts and mining difficulty**, analyzing scalability, load balancing, runtime, and communication overhead.

Span-Aware Cross-Utterance Model (SACUM)

Team of 3

PyTorch, DeBERTa-v3-large, FLAN-T5, HuggingFace | Natural Language Processing

- Developed a modular architecture for **Emotion-Cause Pair Extraction (ECPE)** from multi-turn conversational datasets using DeBERTa-v3-large with speaker and turn-position representations.
- Incorporated **causal cross-utterance attention** with temporal masking and a memory-efficient factored pair-scoring

mechanism to model relationships between emotion and cause spans while controlling GPU memory usage.

- Built an emotion-anchored summarization pipeline using **FLAN-T5**, generating conversational narratives conditioned on extracted emotion-cause pairs and evaluating outputs using **BERTScore and UniEval**.
- Integrated configurable training and evaluation workflows with **automatic mixed precision, dynamic batching, checkpointing, and command-line configuration**.

Congestion-Aware AODV (CA-AODV) **Team of 6**
C++, NS-3 | Computer Networks

- Extended the **AODV routing protocol in NS-3** with congestion-aware route selection using queue occupancy and composite routing metrics for dynamic network conditions.
- Implemented and evaluated enhanced routing strategies incorporating **ETX, congestion-aware path costs, hysteresis-based route switching, and TTL-based route discovery optimization**.
- Compared CA-AODV against baseline AODV across network scenarios using **Packet Delivery Ratio, latency, routing stability, and load-balancing** metrics.

SKILLS

Languages : C, C++, Python, JavaScript
Machine Learning: Supervised and Unsupervised Learning, Classification Algorithms.
Areas of Interest : Machine Learning, Operating Systems, Web Development
Tools & Platforms : Git, Github, Linux (Ubuntu , Fedora), Vs Code, Latex, Solid Edge

RELEVANT COURSES

- **Computer Science:** Introduction to Programming, Data Structures, Algorithms, Software Development Fundamentals, Theory of Computation, Operating Systems, DBMS, Computer Architecture, Compiler Design, Computer Networks, Distributed Computing, Software Engineering, Natural Language Processing
- **Mathematics:** Discrete Mathematics, Calculus, Probability and Random Variables

POSITIONS OF RESPONSIBILITY

OCS Internship Coordinator : Office of Career Services, IIT Hyderabad May'25 - Apr'26
• Assisting in internship process coordination, outreach, and student-industry communication.

Machine Learning Coordinator : Tinkerer's Lab, IIT Hyderabad May'25 - Apr'26
• Organizing hands-on ML workshops and mentoring peers in ML model development and deployment.

Web Coordinator : Ek Bharat Shrestha Bharat (EBSB), IIT Hyderabad Aug'24 - Apr'25
• Maintained the official EBSB website, showcasing cultural events and multimedia galleries. Enabled real-time updates and streamlined event communication across campus.

EXTRACURRICULAR

- **Cricket Team Member, IITH Hostel League :** Represented hostel in inter-hostel cricket tournaments; contributed to team strategy and coordination.
- **Volunteer, NSS (National Service Scheme), IITH :** Participated in organizing blood donation camps, plantation drives, and literacy awareness initiatives.
- **Personal Interests :** Cricket, mobile gaming, spiritual reading, web series enthusiast

EXPERIENCE

Adobe — Content Intelligence Team May 2026 – July 2026
Machine Learning Engineering Intern

- Developed an end-to-end **email layout featurization pipeline** to extract structural intelligence from marketing email HTML for Adobe GenStudio's Content Intelligence platform.
- Rendered email HTML using **Playwright/Chromium** and combined DOM-derived geometry with **Gemini-based semantic annotation** to identify structural elements and hierarchical relationships.
- Designed **canonical layout representations** to preserve spatial structure while reducing sensitivity to textual and visual content, enabling comparison of structurally similar email layouts.
- Evaluated **DINOv2 and Gemini Multimodal embeddings** with multiple clustering approaches to discover recurring email layout families across different representations.
- Implemented deterministic extraction of **22 structural features** covering layout composition, headers, images, grids, CTA placement, and content structure; validated the pipeline on **two independent datasets**, obtaining coherent layout clusters and interpretable features.

COMPETITIVE PROGRAMMING

- **Codeforces:** 1400+ rating
- **LeetCode:** 500+ problems solved